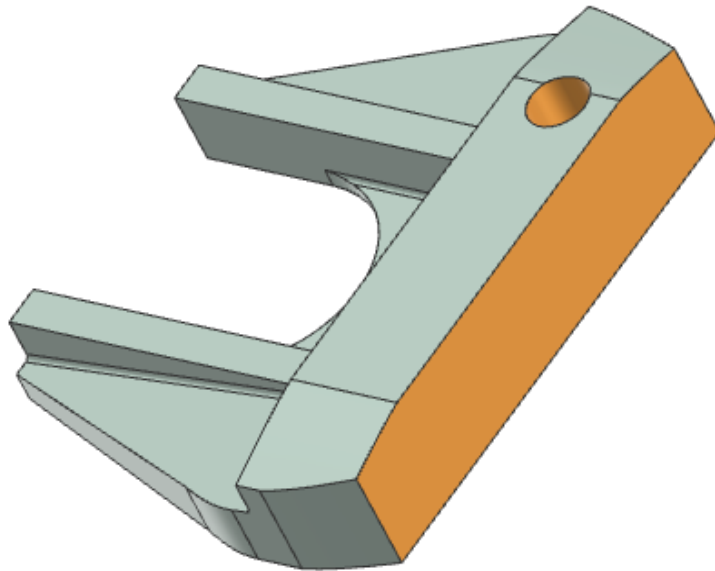


Term Project Final Report

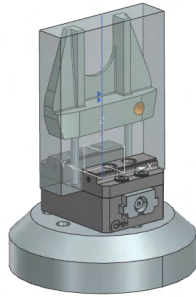
Custom CNC Golf Putter



MFGE 413

Dustin Leung

Project Summary

Project Description	CNC Custom Golf Putter Head	
Project Market	Hobbyist Golfers	
Project Value	Allows for Players to Have Custom Personalized Equipment	
Annual Output	5 Parts Per Day * 260 Working Days = 1300 Parts	
Material Choice	6061 Aluminum Alloy	
Machine Choice	DMG MORI DMU 40	
Fixturing Method	Milling Machine Vise + 2 Tubes and Leftover Blank	
Tooling	Flat End Mill Bull-Nose End Mill Ball End Mill Hole End Mill	¾ in ⅝ in with 0.04in corner radius ⅜ in ¼ in
Time Per Part & Cost Per Part	~ 90 Minutes Per Part	1 Year: \$430 Per Part 3 Year: \$230 Per Part 5 Year: \$190 Per Part
Sale Price & Annual Revenue	\$350 per part	1 Year: -\$168k 2 Year: \$97k 3 Year: \$145k
Sustainability Impacts	Pros: ● Job for Local Economy ● Happiness to Customer	Cons: ● Noise Pollution ● Energy and CO2 Emissions

Introduction

The purpose of this report is to present the development of a complete manufacturing workflow for a custom-designed product using CNC machining. This project demonstrates the entire process of bringing a machined component from concept to production, including product selection, CAD modeling, CAM programming, and fixture design. By progressing through each stage of the manufacturing pipeline, the project highlights the practical considerations involved in machining a real component—such as tool selection, stock preparation, workholding strategies, and toolpath generation.

In addition to the technical manufacturing process, this project includes an evaluation of the product from societal, sustainability, and economic perspectives. Modern engineering practices require more than just functional design; they demand consideration of environmental impacts, long-term resource use, manufacturing efficiency, and cost effectiveness. This report therefore examines how CNC machining influences material waste, energy consumption, production scalability, and the overall economic feasibility of manufacturing the chosen product.

Project Description & Value

Project Description

The product selected for this project is a golf putter, a precision club used by golfers to roll the ball across the green over short distances. A model of the final putter design is shown in Figure 1. Golf putters typically consist of three main components: the grip, the shaft, and the putter head. For this project, the focus is on the putter head, as it is the component most directly influenced by CNC machining.



Figure 1: Putter in Action

The putter head requires precise geometry, tight tolerances, and high-quality surface finishes—areas where CNC machining provides significant value. CNC processes allow for accurate control of complex contours, consistent reproduction of features, and the creation of fine details such as alignment guides and weight-adjustment pockets. Because performance on the green depends heavily on balance, mass distribution, and face flatness, CNC machining is an ideal manufacturing method to meet these demands.

Project Value

Beyond technical performance, the value of the putter also lies in its ability to provide hobbyist golfers with customized equipment. CNC machining enables small-batch or one-off production runs, making it possible for users to personalize attributes such as weight, geometry, aesthetics, and alignment features. This level of customization is difficult or uneconomical with traditional mass-manufacturing processes, but CNC machining makes personalized equipment both accessible and repeatable. As a result, the putter offers both functional precision and individualized value, enhancing the golfing experience for non-professional users.

Annual Output

Annual Output Calculation

The annual output for the CNC-machined putter head is based on the planned production rate and the number of available working days in a year. For this project, we assume a small-scale manufacturing environment intended for custom or low-volume production.

After evaluating the machining operations required for the putter head—including roughing, finishing, contouring, and inspection—the team determined that it is practical to manufacture 5 putter heads per working day. This output accounts for machining time, fixturing, part handling, tool changes, and quality checks.

With an estimated 260 working days per year, and a predicted 5 parts being made daily, the resulting annual output is 1300 parts.

$$5 \text{ Parts Per Day} \times 260 \text{ Working Days} = 1300 \text{ Parts Total}$$

This production rate aligns well with the goals of producing custom and small-batch putters for hobbyist golfers. CNC machining is especially suitable for this scale because it allows for repeatability, precision, and customization without requiring the high upfront costs associated with mass-production tooling. The estimated annual output of 1300 parts therefore represents a realistic and efficient manufacturing capability for this type of product.

Market Considerations for Annual Output

While the production capacity of 1300 putter heads per year demonstrates the feasibility of small-scale CNC manufacturing, the market demand for golf putters presents several challenges. Golf clubs, including putters, are not high-frequency purchases. Most golfers buy a putter infrequently, often keeping the same one for several years. As a result, even well-established manufacturers experience relatively slow turnover compared to industries with consumable or frequently replaced products.

Additionally, the target market for this project is primarily the hobbyist golf community, which tends to be smaller and more niche compared to the broader athletic equipment market. Hobbyist golfers may value customization and unique designs, but the overall customer base is limited. This places natural constraints on sales volume and suggests that large-scale mass production is neither necessary nor economically optimal.

CAD Model & Material Choice

CAD Model

To begin the design phase of the project, a putter head geometry was selected that would demonstrate clear value when manufactured using CNC machining. The initial model was sourced from GrabCAD [1], providing a solid foundation for establishing toolpaths, machining strategies, and fixturing concepts. In a real customer-focused scenario, however, the design process would involve creating a fully custom putter head or modifying an existing geometry based on user preferences.

As shown in Figure 2, the back half of the putter head offers substantial opportunities for customization. Features such as weight pockets, alignment marks, surface contours, and aesthetic details can be modified to suit individual player preference or performance needs. CNC machining is particularly advantageous for these types of features, as it enables precise modifications, rapid iteration, and low-volume personalized production without the need for dedicated molds or casting tools.

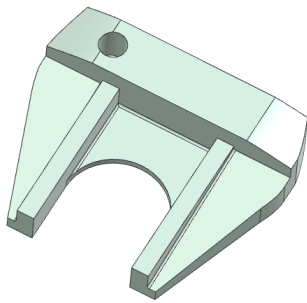


Figure 2: Putter Head Back Half

Material Choice

The putter head for this project is made from aluminum. Aluminum was selected because it is easy to machine, lightweight, and commonly used in modern putter designs. Its low density allows for flexible weight distribution, and it produces smooth finishes with short machining times. This makes aluminum ideal for CNC machining and for creating customized features that hobbyist golfers value. As seen in Fig. 3, an estimated cost for a billet that could create 2 parts is around \$70. This gives us a material cost per part around \$35.

Another material option considered was titanium. Titanium offers excellent strength and durability, but it is significantly more difficult to machine, increases tool wear, and requires longer machining times. For a small-scale, customizable production environment, aluminum provides a more practical and cost-effective choice.

6061 Aluminum Plate

Size:	2	IN.
Width:	5.0000	IN.
Length:	6.0000	IN.
Quantity:	2	Buy More and Save! 3 for \$70.28 each
Price Each: \$ 71.28		

Figure 3: Cost Of Material

Below are the boundary conditions that were considered when selecting the material, as well as the competitive advantages that our selected material gives us.

Boundary Conditions	Competitive Advantage
• Cost of materials • Material properties • Machinability • Accessibility	• Aluminum machinability • Corrosive resistance • Easy to source aluminum • Lower environmental impact

Machine Choice

Machine Choice

For the manufacturing of the golf putter head, the DMG MORI DMU 40 was selected as seen in figure 3. This machine is a compact yet highly capable 5-axis CNC mill, offering the precision and flexibility needed to produce the complex geometry of the putter head. Its multi-axis capabilities allow for efficient machining of angled surfaces, contours, and pockets without requiring multiple setups, improving both accuracy and productivity.



Figure 4: DMG MORI DMU 40

To select the appropriate CNC machine for this project, we compared three 5-axis machining centers: the Haas UMC-400SS, the DMG MORI DMU 40, and the OKUMA M460V-5AX. After evaluating factors such as rigidity, precision, available travel, and suitability for small complex parts, we selected the DMG MORI DMU 40. This machine offers the accuracy and flexibility needed for producing a putter head, while providing reliable 5-axis capability for both orientations required in our machining process. The predicted cost of the DMU 40 we are planning on purchasing is \$400,000.

Below is a table comparing the considered machine options:

Machine Options	Cost	Ranking
Haas UMC-400SS [2]	Starting \$105,000	2nd
DMG MORI DMU 40 [3]	Starting \$250,000	1st
OKUMA M460V-5AX [4]	Starting \$150,000	3rd

Below are some of the boundary conditions we considered when choosing our machine, as well as the competitive advantages our machine choice gives us compared to competitors.

Boundary Conditions

- Table Size
- Spindle Speed
- Tool Changing Capability
- User Interface
- Cost
- Accuracy
- Help Network

Competitive Advantage

- 21.7in x 17.7in x 15.5in
- Cutting Speed - 12000 to 20000 RPM
- SIEMENS Integration
- Palletization and Automatization Potential
- 24/7 Help Network
- 5-Ax Machining
- Easy To Integrate Upgrades From Vendor

Fixturing Plan & Tooling

Fixturing Plan

Fixturing is essential in CNC machining because it determines how securely and accurately a part is held during cutting. A proper fixture keeps the workpiece stable, ensures alignment between operations, and provides repeatable positioning. For a putter head with curved surfaces and varying geometry, effective fixturing is critical for achieving precision and efficiency. Since machining accuracy depends heavily on how the part is held, selecting the right fixture strategy is a key part of this project's manufacturing plan.

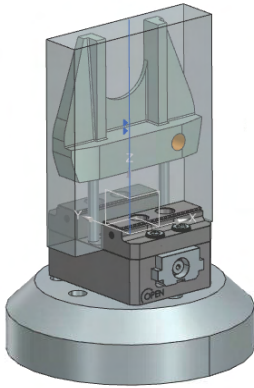


Figure 5: Part In Fixture

For machining the putter head on a 5-axis machine, we selected fixturing methods that fully constrain all six degrees of freedom (three translational and three rotational), ensuring the part remains rigid, stable, and precisely located during each operation.

1. 2513N1 Compact Milling-Machine Vise [5]
2. 2549N7 2 High Riser for Dovetail Milling-Machine Vise [6]
3. 2 Tubes connection to leftover blank in vise

This combination of fixturing choices enables two consecutive operations to be completed entirely within the 5-axis machine setup, minimizing operator involvement and reducing overall handling time. By allowing the putter head to be fully machined in a single, continuous workflow, we improve efficiency, consistency, and throughput for small-batch production.

Below are some of the boundary conditions considered when selecting our fixturing, and the competitive advantages that our choices give us.

Boundary Conditions

- Cost of fixturing
- Control over 6 degrees of freedom
- Complexity of part
- Custom or off the shelf fixturing
- Work to to implement fixturing

Competitive Advantage

- Off the shelf fixturing makes it cheaper
- Single run of CNC Machine
- Low variability between setups
- Set up for automation and palletization
- High cycle potential

Tooling

For our tooling selection, we chose tools that support roughing, surface finishing, and hole-making operations. A serrated carbide roughing end mill is used for fast material removal with lower cutting forces. A solid carbide finishing end mill provides the accuracy and surface quality needed on the final contours. For the hole feature, a carbide drill creates the initial hole, followed by a stub-length end mill to refine its geometry.

Tool	Tool Sizing	Operation	Cutting Speed
Flat End Mill	3/4 in	Roughing Mill	15000 RPM
Bull-Nose End Mill	3/8 in with 0.04in corner radius	Surface Mill	15000 RPM
Ball End Mill	1/8 in	Surface Mill	15000 RPM
Hole End Mill	1/4 in	Hole Mill	15000 RPM

Figure 5: Table of Tools Used for Part

Software, Operations & Milling Results

Software

The software we selected for this project is Siemens NX, chosen primarily for its accessibility, strong CAD/CAM integration, and advanced toolpath generation capabilities. While NX best fits our workflow, we also evaluated several other industry-standard manufacturing software packages before making our decision.

Software	Cost	Ranking
Siemens NX [7]	\$11,600.00	1
Solidworks [8]	\$2,820.00	2
Fusion [9]	\$2,040.00	3

Figure 6: Comparison Of CAD Softwares

Milling Operations

Several machining operations are required to produce the final putter head. First, a roughing operation is used to remove excess material from the billet quickly and efficiently, establishing the basic geometry of the part. This is followed by finishing and surfacing operations, which refine the contours and create the smooth, precise surfaces needed for both performance and aesthetics. Finally, a single hole-making operation is performed to create the attachment point for the putter shaft. The roughing and surface-milling toolpaths used in this workflow can be seen in Figure 5. Together, these operations provide a complete workflow from raw material to a fully machined putter head.



Figure 7: Roughing Mill Tool Path (Left) & Surface Mill Tool Path (Right)

Results of Milling Operations

Using the roughing and finishing strategies described above, we apply this machining cycle to both the top and bottom orientations of the putter head. After completing these two operations, the part is fully machined with the exception of the two remaining fixturing tabs used to secure the workpiece during cutting. These tabs are removed in a secondary step after the part is taken out of the machine, leaving the final putter head ready for any post-processing or finishing steps.



Figure 8: IPW after Bottom Operation (Left) & IPW after Top Operation (Right)

Operation Specifics & Cycle Time

All Programs

By choosing to work with a 5-axis machine, we gain the competitive advantage of completing the entire manufacturing process in a single setup orientation. Despite this, a detailed toolpath still needed to be created, as shown in Fig. 7. As seen in Fig. 9, the listed programs generated in Siemens NX will be used to run the CNC machine and produce the part.

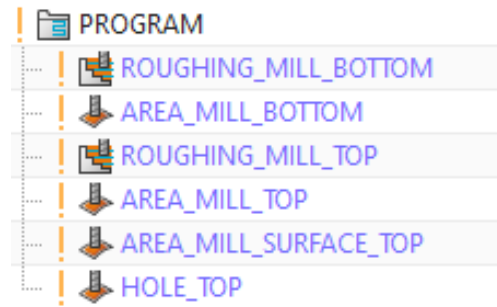


Figure 9: Operations to Complete Part

Cycle Time

As expected, our cycle time is driven primarily by the surface-finishing operations. The small-diameter tools required for these features significantly increase machining time compared to the roughing operations. In particular, the 1/8-inch ball end mill used for detailed surface milling must take many closely spaced passes to achieve the required precision and surface quality. This fine-step machining is essential for the geometry of the putter head but naturally results in longer cycle times. Fig. 10 has all of the programs as listed before along with their associated cut time and length

Title	Time	Length
NC_PROGRAM	00:19:54	3334.9
Unused Items	00:00:00	0.0
PROGRAM	00:19:54	3334.9
ROUGHING_MILL_BOTTOM	00:02:14	344.2
AREA_MILL_BOTTOM	00:01:05	164.6
ROUGHING_MILL_TOP	00:02:11	324.8
AREA_MILL_TOP	00:05:54	1040.2
AREA_MILL_SURFACE_TOP	00:08:04	1398.1
HOLE_TOP	00:00:25	63.1

Figure 10: Time and Length of Each Milling Operation

Below are some boundary conditions considered when creating these tool paths and operations, as well as the competitive advantage we have with our setup

Boundary Conditions

- Tooling Material
- Tooling Size
- Type of Fixturing
- Machine Specifications
- Stock Size

Competitive Advantage

- One Overall Operation - 5-Axis
- Palletization Potential
- Quick Cycle Time
- Customization Without Re-engineering
- Digital Twin Simulation

Pre-Process & Post-Process Work

Pre-Process

To complete the entire part, several steps must be addressed before and after the CNC machining process. One of the primary pre-machining tasks is preparing the raw material. Since we are purchasing a 2-inch aluminum plate, it must be cut down into two 1-inch blanks before fixturing and machining. A bandsaw will be used to split the stock to the required thickness, ensuring that the material is properly sized for the CNC operations that follow.

The part will also require machine setup before machining can begin. This includes installing the appropriate fixtures and tools, as well as transferring the G-code generated in our NX CAM environment to the CNC machine. Additional time is needed to verify offsets, alignment, and toolpaths to ensure the program will run safely and accurately.



Figure 11: Bandsaw for Blank Production

Post-Process

Once the part is removed from the CNC machine, the remaining fixturing features must be removed. The two support tubes used to secure the putter head during machining will be milled off in a secondary operation. This will be done on a manual or secondary milling machine using a custom fixture that ensures the putter head can be aligned consistently along its front face for an even, flat cleanup.

Although the part should come off the CNC machine with relatively high quality, one of the most important aspects of a golf putter is its polished, reflective finish. This “new” shine significantly influences perceived value and is a key selling point in the golf equipment market. To achieve this, we will spend additional time deburring all edges and polishing the surfaces to ensure a smooth, visually appealing final product that aligns with customer expectations and supports our pricing strategy.

At the end of the process, we will also inspect the part to ensure it meets the highest quality standards. Maintaining consistent quality is essential for building and preserving our reputation; even small defects can undermine customer trust and damage credibility. Thorough inspection helps ensure that every putter leaving our shop reflects the level of craftsmanship expected from a reputable manufacturer.



Figure 12: Expected Shine of Product

Time, Costs, & Sale Price

Time To Make One Part

To estimate our total production time, we divided the workflow into three key stages. First, the pre-process stage includes preparing the aluminum billet by cutting it to the required blank size and completing all necessary machine setup. Next, fixturing and CNC machining—described in detail above—cover the roughing, finishing, and hole-making operations needed to fully shape the putter head. Finally, the post-process stage involves deburring and polishing the part, with the final polish playing an important role in achieving the high-quality appearance expected from a premium golf putter.

Pre-Process - 18 Min

1. Billet to Blank - 10 Min
2. Machine Setup - 5 Min
3. CAM G-Code - 3 Min

Fixture & CNC Machining - 28 Min

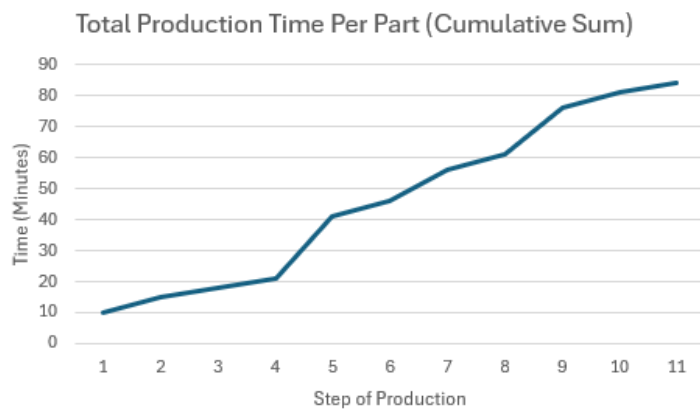
4. Fixturing - 3 Min
5. Milling - 20 Min
6. De-Fixturing - 5 Min

Post Process - 38 Min

7. Remove Tubes - 10 Min
8. Deburr - 5 Min
9. Polishing - 15 Min
10. Cleaning - 5 Min
11. Inspecting - 3 Min

TOTAL ESTIMATED TIME - 84 Min

Figure 13: Estimated Production Time



Cost To Make One Part

The cost analysis of an individual part requires consideration of multiple contributing factors. For this project, all relevant factors listed in the adjacent table were included. A detailed evaluation of these costs and their impact on overall revenue will be presented in the Annual Revenue section later in this report. The associated sources for finding some of these will also be provided. For certain estimates, ChatGPT was used to provide guidance, as I did not yet have sufficient personal experience to make fully informed calculations.

1	Materials	[10]
2	Direct Labor	[12]
3	CNC Machine	[3]
4	Tooling & Fixturing	[5] [6]
5	Consumable	[12]
6	Building	[11]
7	Shipping	[12]
8	Energy	[12]

Figure 14: Important Consideration for Cost of Part

Sale Price

For the sale price of our finished product, a review of similar golf putters on the market reveals a price range of approximately \$250–\$600. Products at the higher end of this range typically offer additional customizations—such as interchangeable weight screws—that provide a competitive advantage and justify a higher price point. Considering these market conditions and applying the common rule of thumb of pricing a product at roughly ten times its material cost, we determined that a sale price of \$350 is both reasonable and competitive for our design.

Annual Revenue

Annual Revenue Estimation

For the annual revenue analysis, I evaluated 1-year, 3-year, and 5-year forecasts to provide a short-, medium-, and long-term view of the business. One of the most significant differences between these projections is the amortization of the major upfront purchase—the CNC machine—which is estimated at \$400,000. While a previous section outlined the cost to produce a single part, the business must also account for broader overhead expenses such as rent, utilities, tooling wear, and labor. These additional costs are reflected in the financial breakdowns shown in the figures on this page.

As shown in the analysis, extending the amortization period increases theoretical profitability because the large upfront machine cost is distributed over more years. However, longer-term forecasts naturally come with greater uncertainty, as unexpected expenses become more likely—whether from equipment maintenance, tool replacement, or changes in operational costs. Therefore, while the projections indicate potential profitability, it is not guaranteed.

Economy Analysis for 1 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 51.20
CNC Machine	\$ 307.69
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 15.38
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ (80.57)
Gross Profit Percentage	-23%
General Overhead Expenses	\$/Unit
Computers	\$ 3.85
Software	\$ 8.85
Non-direct Labor	\$ 25.60
Advertising	\$ 6.15
Insurance	\$ 3.85
Net Profit	\$/Unit
Profit	\$ (128.86)
Total Profit	\$/Year
Total Profit	\$ (167,524.00)

Figure 15: 1 Year Forecast

Economy Analysis for 3 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 53.57
CNC Machine	\$ 102.56
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 16.10
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ 121.47
Gross Profit Percentage	34.71%
General Overhead Expenses	\$/Unit
Computers	\$ 1.28
Software	\$ 8.85
Non-direct Labor	\$ 26.79
Advertising	\$ 6.44
Insurance	\$ 4.02
Net Profit	\$/Unit
Profit	\$ 74.09
Total Profit	\$/Year
Total Profit	\$ 96,317.07

Figure 16: 3 Year Forecast

Economy Analysis for 5 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 55.28
CNC Machine	\$ 61.54
Tooling & Fixturing	\$ 3.60
Consumable	\$ 3.85
Building	\$ 16.61
Shipping	\$ 13.08
Energy	\$ 0.77
Gross Profit	\$/Unit
Profit	\$ 160.28
Gross Profit Percentage	46%
General Overhead Expenses	\$/Unit
Computers	\$ 0.77
Software	\$ 8.85
Non-direct Labor	\$ 27.64
Advertising	\$ 6.64
Insurance	\$ 4.15
Net Profit	\$/Unit
Profit	\$ 112.23
Total Profit	\$/Year
Total Profit	\$ 145,897.02

Figure 17: 5 Year Forecast

As seen above, when looking at a short term forecast (see Fig. 15), the business is not profitable at all and will lead to large money losses. However, as we forecast longer and spread out the larger cost of the business, we start to see profitability in Fig. 16 & 17. This in many ways proves to us our predicted sales price will match well with our planned costs of manufacturing to create this product.

Annual Revenue Assuming Ideal Market Conditions

Annual Revenue Estimation For Ideal Conditions

If we can assume lights off manufacturing, and a constant supply of customers, we can create an estimation for an automated manufacturing line of putters. In this case, we must purchase some upgrades for our CNC, purchase a robot to perform the swapping of pallets, and buy more fixturing. This is reflected in the analysis figures on this page. For our robot we estimated a price of around \$100,000 [13].

The number of parts also increases a bunch, so instead of around 5 parts made daily, we go to 60, and the working days are 365 days. Thus, we expect to produce 22,000 parts yearly.

Economy Analysis for 1 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 3.03
CNC Machine	\$ 36.36
Robot	\$ 4.55
Tooling & Fixturing	\$ 5.65
Consumable	\$ 0.23
Building	\$ 0.91
Shipping	\$ 0.77
Energy	\$ 0.05
Gross Profit	\$/Unit
Profit	\$ 263.46
Gross Profit Percentage	75%
General Overhead Expenses	\$/Unit
Computers	\$ 0.23
Software	\$ 0.52
Non-direct Labor	\$ 1.51
Advertising	\$ 0.36
Insurance	\$ 0.23
Net Profit	\$/Unit
Profit	\$ 260.61
Total Profit	\$/Year
Total Profit	\$ 5,733,412.00

Figure 18: 1 Year Forecast

Economy Analysis for 3 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 3.17
CNC Machine	\$ 12.12
Robot	\$ 1.52
Tooling & Fixturing	\$ 5.65
Consumable	\$ 0.23
Building	\$ 0.95
Shipping	\$ 0.77
Energy	\$ 0.05
Gross Profit	\$/Unit
Profit	\$ 290.55
Gross Profit Percentage	83.02%
General Overhead Expenses	\$/Unit
Computers	\$ 0.08
Software	\$ 0.52
Non-direct Labor	\$ 1.58
Advertising	\$ 0.38
Insurance	\$ 0.24
Net Profit	\$/Unit
Profit	\$ 287.75
Total Profit	\$/Year
Total Profit	\$ 6,330,586.41

Figure 19: 3 Year Forecast

Economy Analysis for 5 Year Forecast	
Sales Price	\$/Unit
CNC Custom Golf Putter	\$ 350.00
Cost of Goods Sold	\$/Unit
Materials	\$ 35.00
Direct Labor	\$ 3.27
CNC Machine	\$ 7.27
Robot	\$ 0.91
Tooling & Fixturing	\$ 0.21
Consumable	\$ 0.23
Building	\$ 0.98
Shipping	\$ 0.77
Energy	\$ 0.05
Gross Profit	\$/Unit
Profit	\$ 301.31
Gross Profit Percentage	86%
General Overhead Expenses	\$/Unit
Computers	\$ 0.05
Software	\$ 0.52
Non-direct Labor	\$ 1.63
Advertising	\$ 0.39
Insurance	\$ 0.25
Net Profit	\$/Unit
Profit	\$ 298.47
Total Profit	\$/Year
Total Profit	\$ 6,566,397.02

Figure 20: 5 Year Forecast

Similarly to our more realistic calculated numbers, in the short term, you are very unlikely to create a profitable business due to the high upfront costs of the machine, robot, and all the fixturing you need to reach automation. However, as these rough calculations show, you would be able to get a lot more net profit as you are creating and selling more parts.

Sustainability Impacts

Society

For society, we can view our project as bringing positive contributions because it supports the local economy by creating a job opportunity. We are also providing someone with fulfilling work that can help strengthen their self-esteem. In addition, our product should enhance customer satisfaction and bring genuine happiness to the people who use it. Importantly we are targeting a net positive for the maslow’s hierarchy of needs.



Figure 21: Maslow’s Hierarchy of Needs

There are also potential drawbacks to consider. Our product may have negative environmental impacts, which will be discussed in more detail in the environmental section. Additionally, the operation of our business could contribute to local noise pollution, especially if located near residential areas. However, we are committed to mitigating these issues through responsible planning, sustainable practices, and thoughtful facility placement to minimize any negative effects on the community.

Economy

Much of the economic analysis builds directly on the annual revenue calculations discussed above. Over longer time horizons, profitability is expected to improve as major equipment costs are amortized, lowering the effective yearly expenditure. However, it is important to acknowledge that estimates become less reliable the further out they extend. Unexpected expenses—such as increases in operating costs, general living expenses, and wear or maintenance requirements for machinery—can significantly affect long-term projections.

Environmental

An important consideration when manufacturing any part is its environmental impact. Every stage of the process—from raw material extraction to machining and waste disposal—contributes to the overall footprint. Because these impacts can be difficult to quantify accurately, we used Ansys Granta to support our analysis and provide more reliable estimates of the material and process-related environmental effects. Below, you can see some estimations based on what we plan on doing (green), versus choosing different materials and other methods of transportation and packaging (yellow). Fig. 20, has the listed differences.

	Planned Part	Tested Changes
Material	Aluminum	Titanium
Packaging	Cardboard	Polystyrene
Travel	Seacraft	Aircraft

Figure 22: Table of Differences

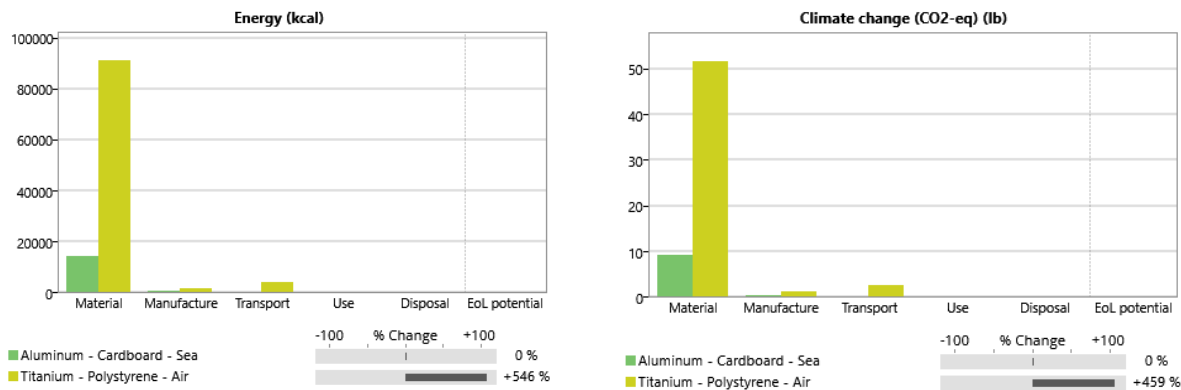


Figure 23: Energy and CO2 Impact of Our Production

Boundary Conditions & Competitive Advantage

Statement

A key part of this project was evaluating the boundary conditions and competitive advantages of our product. Boundary conditions define the practical limitations and requirements we must consider when making design and purchasing decisions, while competitive advantages determine how those choices help the product stand out in the market. Although these ideas were discussed throughout the report, the following section provides a comprehensive summary of the most important points.

Boundary Conditions of Project

Product

1. Value
2. Product Complexity
3. CNC Feasibility

Material

1. Material Cost
2. Sourcing Ability
3. Machinability

Machine

1. Number of Axes
2. Purchase Cost
3. Workspace Size
4. Spindle Speed
5. User Interface

Fixturing

1. 6 Degrees of Freedom
2. Fixturing Cost
3. Repeatability
4. Palletization Potential

Market

1. Purchasing Frequency
2. Brand Reputation
3. Competitive Pricing

Environmental

1. Transportation
2. Maslow's Laws
3. Energy Consumption
4. Location

Competitive Advantages of Project

Product

1. High Perceived Value
2. CNC Machine Preferred
3. Fully Customizable

Material

1. Easy to Source
2. High Machinability
3. Strength to Weight Ratio
4. Non-Toxic

Machine

1. 5-Axis Machine
2. Low Operator Intervention
3. High Precision and Repeatability
4. Surface Quality Finish
5. Cutting Speed
6. Support from Vendor
7. CAM Compatibility
8. Upgrades Available

Fixturing

1. Palletization
2. One Set Up
3. Easy to Source
4. Flexibility for Future Designs
5. Scalability

Market

1. Personalization
2. Easy to Swap Designs
3. Expansion Potential
4. High-Quality Machining
5. Opportunity to Create Brand Identity

Environmental

1. Recyclable material
2. Local Job for Economy
3. Customer Satisfaction
4. Personal Self Worth

Conclusion

Project Conclusion

Manufacturing a golf putter head on a CNC machine—under favorable market conditions—has the potential to be a viable and profitable business. The part carries a high retail value, is relatively straightforward to machine once the process is established, and can offer strong margins due to its low material cost and premium pricing. In theory, this combination presents an appealing opportunity for a small-scale manufacturing venture.

However, as this report demonstrates, there are numerous boundary conditions that must be considered before undertaking a CNC-based business. Significant upfront costs for equipment, tooling, and workspace must be amortized over time. The golf putter market is highly competitive and saturated, with low purchase frequency among consumers, which limits potential revenue. Success in such a market depends not only on efficient production but also on establishing a clear competitive advantage—through superior quality, customization options, or innovative designs—compared to peers. Additionally, operational considerations such as fixturing, tooling selection, and process optimization all contribute to the complexity and risk of the venture. Taking these factors into account, while the idea is theoretically profitable, it is not a business model I would realistically pursue in the real world.

Personal Conclusion

This class provided valuable insight into the challenges of creating a successful business. Beyond the technical CAD/CAM work, it became clear that understanding boundary conditions, selecting the right equipment for a competitive advantage, and managing costs are all critical and complex tasks. Developing the financial analysis highlighted how difficult it is to estimate expenses accurately and anticipate hidden costs, which can significantly impact profitability. While my setup demonstrates potential for success, the process made me aware of the many factors that can influence real-world outcomes. This experience has reinforced the importance of holistic planning, critical evaluation of assumptions, and risk awareness—lessons that will be essential in any future engineering or entrepreneurial endeavor.

Sources

Note: ChatGPT was used to support the writing and editing of this document. All CAD and CAM work, as well as the development of the manufacturing process, were completed by the author; AI assistance was used only to supplement the written content unless stated otherwise.

[1] GrabCAD, “Putter,” Available: <https://grabcad.com/library/putter-12>

[2] Haas Automation, “UMC-400SS — Universal Machining Center,” Available: <https://www.haascnc.com/machines/vertical-mills/universal-machine/models/umc-400SS.html>

[3] DMG MORI, “DMU 40 – 5-Axis Milling,” Available: <https://us.dmgmori.com/products/machines/milling/5-axis-milling/dmu/dmu-40>

[4] OKUMA, “GENOS M460V-5AX — 5-Axis Vertical Machining Center,” Available: <https://www.okuma.com/products/genos-m460v-5ax>

[5] McMaster-Carr, “Compact Milling-Machine Vises” Available: <https://www.mcmaster.com/catalog/131/2967/2513N1>

[6] McMaster-Carr, “Dovetail Milling-Machine Vises” Available: <https://www.mcmaster.com/catalog/131/2893/2549N7>

[7] Siemens Digital Industries Software, “NX X CAD Licensing – 50 Tokens,” Available: <https://plm.sw.siemens.com/en-US/nx/products/nx-x-cad-licensing-50-tokens/>

[8] Dassault Systèmes, “SOLIDWORKS Design Cloud Services – Plans & Pricing,” Available: <https://www.solidworks.com/how-to-buy/solidworks-design-cloud-services-plans-pricing>

[9] Autodesk, “Fusion 360 Offerings,” Available: <https://www.autodesk.com/campaigns/fusion-360/offerings>

[10] Midwest Steel & Aluminum, “6061 Aluminum Plate,” Available: <https://www.midweststeelsupply.com/store/6061aluminumplate>

[11] LoopNet, “27630 Llewellyn Rd, Corvallis, OR,” Available: <https://www.loopnet.com/Listing/27630-Llewellyn-Rd-Corvallis-OR/33858282/>

[12] OpenAI, “ChatGPT (GPT-5),” AI language model, Available: <https://chat.openai.com/>

[13] Standard Bots, “Robotic Palletizing Cost: A Complete 2024 Guide.” Accessed: Feb. 2025. [Online]. Available: <https://standardbots.com/blog/robotic-palletizing-cost>